

EDS Lab Assignments

Practical 5

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5.1.1 Stacked Plot

Problem Statement:

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Code:

```
import matplotlib.pyplot as plt
import pandas as pd
```

```
# Data for Months and Temperature for three cities
```

```
data = {
    'Month': ['January', 'February', 'March', 'April', 'May', 'June', 'July', 'August', 'September',
    'October', 'November', 'December'],
    'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
    'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
}
```

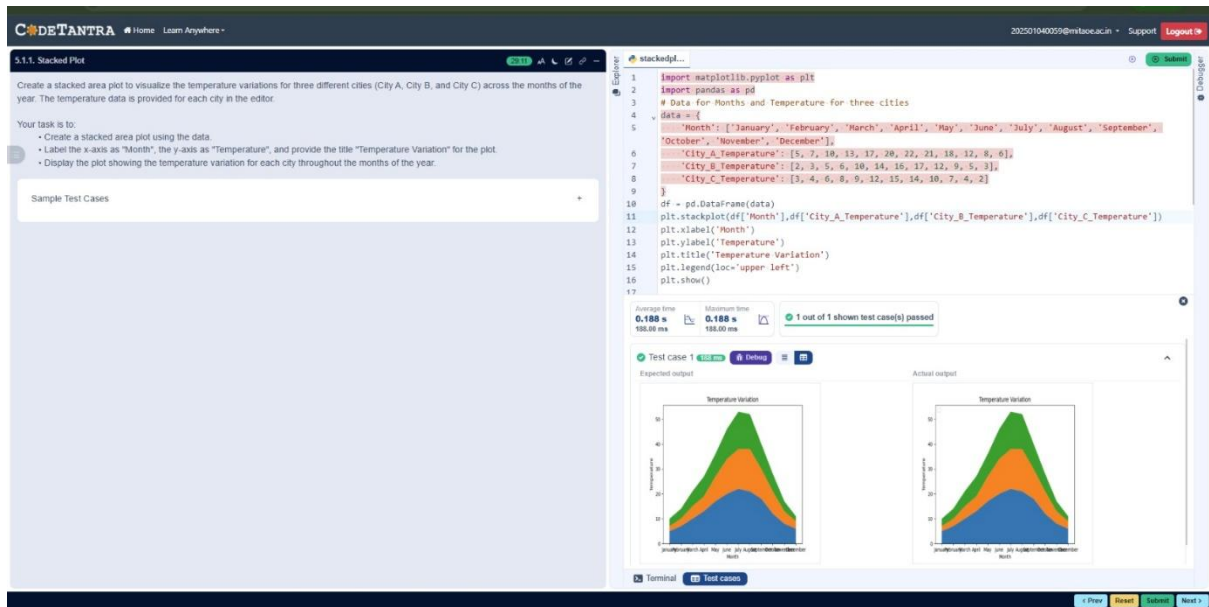
```
# Write your code...
```

```
df = pd.DataFrame(data)
```

```
plt.stackplot(df['Month'], df['City_A_Temperature'], df['City_B_Temperature'],
df['City_C_Temperature'])
plt.xlabel('Month')
plt.ylabel('Temperature')
```

```
plt.title('Temperature Variation')
plt.legend(loc = 'upper left')
plt.show()
```

Output:



5.2.1 Titanic Dataset

Problem Statement:

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the pandas library. It contains the following columns:

- **Pclass:** Passenger class (1 = First, 2 = Second, 3 = Third).
- **Gender:** Gender of the passenger (male/female).
- **Age:** Age of the passenger.
- **Survived:** Survival status (0 = Did not survive, 1 = Survived).
- **Fare:** Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (Pclass).
- Use the color `skyblue` for the bars.
- Title the plot as **"Passenger Class Distribution"**.
- Label the x-axis as **"Pclass"** and the y-axis as **"Count"**.

Pie Chart (Gender Distribution):

- Create a pie chart to display the distribution of male and female passengers.
- Use `lightblue` for males and `lightcoral` for females.
- Include percentages on the slices (use `autopct='%1.1f%%'`).
- Title the plot as **"Gender Distribution"**.

Histogram (Age Distribution):

- Create a histogram to visualize the distribution of passengers' ages.
- Use `lightgreen` for the bars with black edges (`edgecolor = 'black'`).
- Set the number of bins to **8** for the histogram.
- Title the plot as **"Age Distribution"**.
- Label the x-axis as **"Age"** and the y-axis as **"Frequency"**.

Bar Plot (Survival Count):

- Create a bar plot to show the count of passengers who survived and those who did not, based on the `Survived` column.
- Use the colors `lightblue` for survivors (1) and `lightcoral` for non-survivors (0).
- Title the plot as **"Survival Count"**.
- Label the x-axis as **"Survived (0 = No, 1 = Yes)"** and the y-axis as **"Count"**.

Scatter Plot (Fare vs Age):

- Create a scatter plot to visualize the relationship between the Fare and Age of passengers.
- Use orange for the data points.
- Title the plot as "**Fare vs Age**".
- Label the x-axis as "**Age**" and the y-axis as "**Fare**".

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset from the CSV file
df = pd.read_csv('titanic.csv')

# Set up the figure for 5 subplots
fig, axes = plt.subplots(3, 2, figsize=(12, 12))

# Plot 1: Count of passengers by class
axes[0, 0].bar(df['Pclass'].value_counts().index, df['Pclass'].value_counts(),
               color='skyblue')
axes[0, 0].set_title("Passenger Class Distribution")
axes[0, 0].set_xlabel("Pclass")
axes[0, 0].set_ylabel("Count")

# Plot 2: Gender distribution
axes[0, 1].pie(df['Gender'].value_counts(), labels=df['Gender'].value_counts().index,
               autopct='%1.1f%%', colors=['lightblue', 'lightcoral'])
axes[0, 1].set_title("Gender Distribution")

# Plot 3: Age distribution
axes[1, 0].hist(df['Age'].dropna(), bins=8, color='lightgreen', edgecolor='black')
axes[1, 0].set_title("Age Distribution")
axes[1, 0].set_xlabel("Age")
axes[1, 0].set_ylabel("Frequency")

# Plot 4: Survival count
axes[1, 1].bar(df['Survived'].value_counts().index, df['Survived'].value_counts(),
               color=['lightblue', 'lightcoral'])
axes[1, 1].set_title("Survival Count")
axes[1, 1].set_xlabel("Survived (0 = No, 1 = Yes)")
```

```
axes[1, 1].set_ylabel("Count")
```

```
# Plot 5: Fare vs Age
```

```
axes[2, 0].scatter(df['Age'], df['Fare'], color='orange', edgecolors='black')
```

```
axes[2, 0].set_title("Fare vs Age")
```

```
axes[2, 0].set_xlabel("Age")
```

```
axes[2, 0].set_ylabel("Fare")
```

```
plt.tight_layout()
```

```
plt.show()
```

Output:

The screenshot shows the CODETANTRA IDE interface. On the left, a sidebar displays instructions for analyzing the Titanic dataset. The main editor area contains Python code for data analysis and visualization. The code includes imports for pandas and matplotlib, loading the Titanic dataset, and creating five subplots: Passenger Class Distribution (bar chart), Gender Distribution (pie chart), Age Distribution (histogram), Survival Count (bar chart), and Fare vs Age (scatter plot). The code uses various matplotlib functions like bar, pie, hist, and scatter, and includes labels and titles for each plot. The output of the code is not visible in this screenshot.

This screenshot shows the same CODETANTRA IDE interface, but with the output of the code visible. The left sidebar contains instructions for the Titanic dataset analysis. The main editor area shows the Python code. Below the code, a section titled "Test case 1" displays the "Expected output" and "Actual output". The "Expected output" shows five subplots: a bar chart for Passenger Class Distribution, a pie chart for Gender Distribution, a histogram for Age Distribution, a bar chart for Survival Count, and a scatter plot for Fare vs Age. The "Actual output" shows the same five subplots, confirming that the code executed successfully and produced the expected visualizations.

5.2.2 Histogram of Passenger Information of Titanic

Problem Statement:

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Histogram
plt.hist(data['Age'], bins = 30, edgecolor = 'k')
plt.title("Age Distribution")
plt.xlabel('Age')
plt.ylabel("Frequency")
plt.show()
```

Output:

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5.2.2. Histogram of passenger information of Titanic

Write a Python code to plot a histogram for the distribution of the 'Age' column from the 'Titanic' dataset. The histogram should display the frequency of different age ranges with the following specifications:

- Use 30 bins for the histogram.
- Set the edge color of the bars to black (k).
- Label the x-axis as 'Age' and the y-axis as 'Frequency'.
- Add the title "Age Distribution" to the histogram.

The Titanic dataset contains columns as shown below.

Passenger ID	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.25	0	S
2	1	1	Mr. Cumings, Mrs. John Bradley (Florence Briggs "Aunt")	female	38.0	1	0	PC 17599	53.1	0	C
3	1	3	Mr. McKim, Miss. Laura	female	26.0	0	0	STON/O2 3101282	7.92	0	S
4	1	1	Mr. Hays, Mr. William (Hiram) "Uncle" Hays	male	30.0	1	0	110393	53.1	0	S
5	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
6	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
7	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
8	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
9	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
10	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
11	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
12	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
13	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
14	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
15	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
16	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
17	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
18	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
19	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
20	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
21	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
22	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S
23	0	3	Mr. Brown, Mr. Thomas	male	29.0	1	0	110540	5.00	0	S

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1, 0, 3, "Mr. Braund, Mr. Owen Harris", male, 22.0, 1, 0, "A/5 21171", 7.25, 0, "S"
2, 1, 1, "Mrs. Cumings, Mrs. John Bradley (Florence Briggs 'Aunt')", female, 38.0, 1, 0, "PC 17599", 53.1, 0, "C"
3, 1, 3, "Mr. McKim, Miss. Laura", female, 26.0, 0, 0, "STON/O2 3101282", 7.92, 0, "S"
4, 1, 1, "Mr. Hays, Mr. William (Hiram) 'Uncle' Hays", male, 30.0, 1, 0, "110393", 53.1, 0, "S"
5, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
6, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
7, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
8, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
9, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
10, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
11, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
12, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
13, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
14, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
15, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
16, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
17, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
18, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
19, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
20, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
21, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
22, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
23, 0, 3, "Mr. Brown, Mr. Thomas", male, 29.0, 1, 0, "110540", 5.00, 0, "S"
```

Note: Refer to the visible test case for better reference.

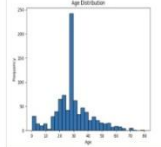
Sample Test Cases

Histogram

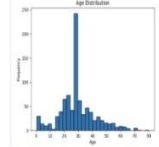
```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.hist(data['Age'], bins=30, edgecolor='k')
18
19 plt.title('Age Distribution')
20 plt.xlabel('Age')
21 plt.ylabel('Frequency')
22 plt.show()
23
```

Test case 1 Pass Debug Test

Expected output



Actual output



Terminal Test Cases

1 Pass 0 Fail 0 Submit Next

5.2.3 Bar Plot of Survival Rate of Passengers

Problem Statement:

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the '**Survived**' column to show the count of survivors (0 = Did not survive, 1 = Survived).
2. Set the chart type to '**bar**'.
3. Add the title "**Survival Count**" to the chart.
4. Label the x-axis as '**Survived**' and the y-axis as '**Count**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival Rate
survival_counts = data['Survived'].value_counts()
survival_counts.plot(kind='bar')
plt.title('Survival Count')
plt.xlabel('Survived')
plt.ylabel('Count')
plt.show()
```


Output:

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5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:
1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived)
2. Set the chart type to 'bar'
3. Add the title 'Survival Count' to the chart.
4. Label the x-axis as 'Survived' and the y-axis as 'Count'.
The Titanic dataset contains columns as shown below.

Passenger id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Brown, Mr. Owen Harris	male	22.0	0	0	5171.7	53.0	5	
2	1	1	Mr. Cumings, Mrs. John Bradley (Florence Briggs Taper)	female	38.1	0	0	3101.0	71.2833	140.5	
3	1	1	Mr. McKim, Miss. Lillian	female	26.0	0	0	5170.0	53.0	5	
4	1	1	Mr. Tremain, Mrs. Jacques Heath (Lily May Peel)	female	35.0	0	0	5170.0	53.0	5	
5	0	3	Mr. Allen, Mr. William Henry	male	35.0	0	0	5170.0	53.0	5	
6	0	3	Mr. Brown, Mr. James	male	28.0	0	0	5171.7	53.0	5	
7	0	1	Mr. McCarthy, Mr. Timothy J.	male	34.0	0	0	5170.0	53.0	5	
8	0	3	Mr. McKim, Mr. George (Edward)	male	22.0	0	0	5170.0	53.0	5	
9	1	1	Mr. Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg)	female	27.0	0	0	5170.0	53.0	5	
10	1	1	Mr. Brown, Mrs. Nicholas (Abigail Achen)	female	31.0	0	0	5171.7	53.0	5	

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,Mr. Brown, Mr. Owen Harris,male,22.0,0,0,5171.7,53.0,5,
2,1,1,Mr. Cumings, Mrs. John Bradley (Florence Briggs Taper),female,38.1,0,0,3101.0,71.2833,140.5,
3,1,1,Mr. McKim, Miss. Lillian,female,26.0,0,0,5170.0,53.0,5,
4,1,1,Mr. Tremain, Mrs. Jacques Heath (Lily May Peel),female,35.0,0,0,5170.0,53.0,5,
5,0,3,Mr. Allen, Mr. William Henry,male,35.0,0,0,5170.0,53.0,5,
6,0,3,Mr. Brown, Mr. James,male,28.0,0,0,5171.7,53.0,5,
7,0,1,Mr. McCarthy, Mr. Timothy J.,male,34.0,0,0,5170.0,53.0,5,
8,0,3,Mr. McKim, Mr. George (Edward),male,22.0,0,0,5170.0,53.0,5,
9,1,1,Mr. Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg),female,27.0,0,0,5170.0,53.0,5,
10,1,1,Mr. Brown, Mrs. Nicholas (Abigail Achen),female,31.0,0,0,5171.7,53.0,5,

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlot.py

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival Rate
17 survival_counts = data['Survived'].value_counts()
18 survival_counts.plot(kind='bar')
19
20 plt.title('Survival Count')
21 plt.xlabel('Survived')
22 plt.ylabel('Count')
23 plt.show()
```

Average time: 0.235 s
Maximum time: 0.235 s
1 out of 1 shown test case(s) passed

Test case 1
Expected output: Survival Count
Actual output: Survival Count

5.2.4 Bar Plot for Survival by Gender

Problem Statement:

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the '**Sex**' column, then use the **value_counts()** function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "**Survival by Gender**" to the chart.
4. Label the x-axis as '**Gender**' and the y-axis as '**Count**'.
5. The legend should indicate '**Not Survived**' and '**Survived**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Gender
survived_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
survived_counts.plot(kind = 'bar', stacked = True)

plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output:

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

- Group the data by the 'Sex' column, then use the `value_counts()` function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.
- Use a stacked bar chart to display the survival counts.
- Add the title "Survival by Gender" to the chart.
- Label the x-axis as 'Gender' and the y-axis as 'Count'.
- The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below.

Passenger Id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Tayer)",female,36,1,0,PC 17599,71.00,S,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2 3101282,7.92,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,310154,8.0,S
6,0,3,"Moran, Mr. James",male,0,0,310177,0,0,S
7,0,1,"McCarthy, Mr. Timothy J.",male,54,0,0,17461,51.00,S,C40,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,104909,21.0,S
9,1,1,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,1,107782,11.1333,S
10,1,2,"Nasser, Mrs. Nicholas (Dionise Nasse)",female,16,1,0,210796,18.00,S,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

BarPlotCI...

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Gender
17 survived_counts = data.groupby('Sex')['Survived'].value_counts().unstack()
18 survived_counts.plot(kind='bar', stacked=True)
19
20 plt.title('Survival by Gender')
21 plt.xlabel('Gender')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

Average time: 0.117 s, Maximum time: 0.117 s, 1 out of 1 shown test case(s) passed

Test case 1: Expected output: Actual output: Both show stacked bar charts for Survival by Gender. The 'Not Survived' series (blue) is significantly higher than the 'Survived' series (orange) for both male and female groups.

Terminal

Test cases

5.2.5 Bar Plot for Survival by Pclass

Problem Statement:

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (**Pclass**), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the **Pclass** column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using **value_counts()**.
2. Use a **stacked bar chart** to display the survival counts.
3. Add the title "**Survival by Pclass**" to the chart.
4. Label the x-axis as '**Pclass**' and the y-axis as '**Count**'.
5. The legend should indicate '**Not Survived**' and '**Survived**'.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Pclass
survival_counts = data.groupby('Pclass')['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind = 'bar', stacked=True)

plt.title('Survival by Pclass')
plt.xlabel('Pclass')
plt.ylabel('Count')

plt.legend(['Not Survived', 'Survived'])
plt.show()
```

Output:

CODETANTRA

Home Learn Anywhere

2025/10/25 8:00:00 AM Logout

2.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Pclass" to the chart.

4. Label the x-axis as "Pclass" and the y-axis as "Count".

5. The legend should indicate "Not Survived" and "Survived".

The Titanic dataset contains columns as shown below.

Passenger ID	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
passengerId,survived,pclass,name,sex,age,sibsp,parch,ticket,fare,cabin,embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,2101,7.25,S
2,1,1,"Goretti, Mrs. John Bradley (Lucenia Briggs Thayer)",female,38,1,0,0,17509,51.0633,C
3,1,1,"Mackenzie, Miss. Kate",female,30,0,0,0,17000,21.021,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,1,13309,53.1,C
5,0,1,"Allen, Mr. William Henry",male,35,0,0,0,17400,8.0,S
6,0,3,"Moran, Mr. James",male,19,0,1,0,8451,5.488,Q
7,0,1,"McCarthy, Mr. Timothy J",male,34,0,0,1,1601,51.8625,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,1,1,0,40859,21.075,S
9,1,1,"Johnson, Mrs. Oscar W (Ellenest Vilhelmine Berg)",female,27,0,2,0,14702,51.1333,S
10,1,1,"Nassery, Mrs. Nicholas (Jahida Rahman)",female,34,1,0,2,17756,50.4708,C
```

Note:

- Refer to the visible test case for better reference.
- Ensure you use the `groupby()` function with `value_counts()` to count the survivors and non-survivors for each `Pclass`.
- Do not manually use `size()` or `unstack()` without `value_counts()`. Use the `value_counts()` method for counting survival status directly.

Sample Test Cases

Editor

BarPlotSt_

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'] = data['Age'].median(), inplace=True
9 data['Embarked'] = data['Embarked'].mode()[0], inplace=True
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male':0, 'female':1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Pclass
17 grouped = pd.crosstab(data['Pclass'], data['Survived'])
18 grouped.plot(kind='bar', stacked=True)
19
20 plt.title('Survival by Pclass')
21 plt.xlabel('Pclass')
22 plt.ylabel('Count')
23 plt.legend(['Not Survived', 'Survived'])
24 plt.show()
```

0.291 s

0.291 s

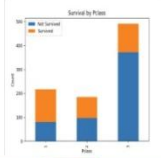
1 out of 1 shown test case(s) passed

Test case 1

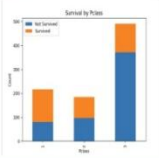
Expected output

Actual output

Survival by Pclass



Survival by Pclass



Terminal

Test cases

Prev Next

5.2.6 Bar Plot for Survival by Embarked

Problem Statement:

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.

The chart should display the following specifications:

1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked** " to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Bar Plot for Survival by Embarked
survival_counts =
    data.groupby('Embarked_Q')['Survived'].value_counts().unstack().fillna(0)

survival_counts.plot(kind = 'bar', stacked = True)
plt.title('Survival by Embarked')
plt.xlabel('Embarked')
plt.ylabel('Count')
```

```
plt.legend(['Not Survived', 'Survived'])  
plt.show()
```

Output:

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset. The chart should display the following specifications:

1. Use the Embarked column to determine the embarkation location. After converting this column into dummy variables (using `pd.get_dummies()`), plot the survival count based on the Embarked, Q column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "Survival by Embarked" to the chart.
4. Label the x-axis as "Embarked" and the y-axis as "Count".
5. Include a legend to distinguish between survivors and non-survivors (label the legend as "Survived" and "Not Survived").

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Brown, Mr. Owen Henry	male	22.0	0	0	10151	7.25	5	Q
2	1	1	Mr. Gage, Mrs. John Bradley (Florence Briggs Thayer)	female	38.1	0	0	17599	71.2833	C85	Q
3	1	3	Mr. McKelvey, Miss. Laine	female	26.0	0	0	106402	7.00	3	Q
4	1	1	Mr. Brown, Mrs. Josephine (Miss. Mrs. Brown)	female	35.0	0	0	113801	51.0	C123	Q
5	0	3	Mr. Allen, Mr. William Henry	male	15.0	0	0	17418	0.49	5	Q
6	0	3	Mr. Brown, Mrs. Josephine	female	38.0	0	0	10151	7.25	5	Q
7	0	3	Mr. Brown, Mrs. Josephine	female	38.0	0	0	10151	7.25	5	Q
8	0	3	Mr. Brown, Mrs. Josephine	female	38.0	0	0	10151	7.25	5	Q
9	0	3	Mr. Brown, Mrs. Josephine	female	38.0	0	0	10151	7.25	5	Q
10	1	1	Mr. Brown, Mrs. Josephine	female	38.0	0	0	10151	7.25	5	Q

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked  
1, 0, 3, Mr. Brown, Mr. Owen Henry, male, 22.0, 0, 0, 10151, 7.25, 5, Q  
2, 1, 1, Mr. Gage, Mrs. John Bradley (Florence Briggs Thayer), female, 38.1, 0, 0, 17599, 71.2833, C85, Q  
3, 1, 3, Mr. McKelvey, Miss. Laine, female, 26.0, 0, 0, 106402, 7.00, 3, Q  
4, 1, 1, Mr. Brown, Mrs. Josephine (Miss. Mrs. Brown), female, 35.0, 0, 0, 113801, 51.0, C123, Q  
5, 0, 3, Mr. Allen, Mr. William Henry, male, 15.0, 0, 0, 17418, 0.49, 5, Q  
6, 0, 3, Mr. Brown, Mrs. Josephine, female, 38.0, 0, 0, 10151, 7.25, 5, Q  
7, 0, 3, Mr. Brown, Mrs. Josephine, female, 38.0, 0, 0, 10151, 7.25, 5, Q  
8, 0, 3, Mr. Brown, Mrs. Josephine, female, 38.0, 0, 0, 10151, 7.25, 5, Q  
9, 0, 3, Mr. Brown, Mrs. Josephine, female, 38.0, 0, 0, 10151, 7.25, 5, Q  
10, 1, 1, Mr. Brown, Mrs. Josephine, female, 38.0, 0, 0, 10151, 7.25, 5, Q
```

Note: Refer to the visible test case for better reference.

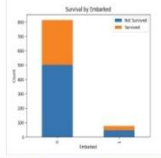
Sample Test Cases

BarPlotC...

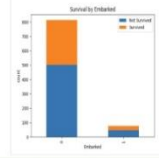
```
1 import pandas as pd  
2 import matplotlib.pyplot as plt  
3  
4 # Load the Titanic dataset  
5 data = pd.read_csv('Titanic-Dataset.csv')  
6  
7 # Data Cleaning  
8 data['Age'].fillna(data['Age'].median(), inplace=True)  
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)  
10 data.drop('Cabin', axis=1, inplace=True)  
11  
12 # Convert categorical features to numeric  
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})  
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)  
15  
16 # Write your code here for Bar Plot for Survival by Embarked  
17 grouped = pd.crosstab(data['Embarked_0'], data['Survived'])  
18 grouped.plot(kind='bar', stacked=True)  
19 plt.xlabel('Embarked')  
20 plt.ylabel('Count')  
21 plt.title('Survival by Embarked')  
22 plt.legend(['Not Survived', 'Survived'])  
23 plt.show()
```

Test case 1

Expected output



Actual output



Terminal

Test cases

1 Prev 2 Next 3 Submit 4 Next 5

5.2.7 Box Plot for Age Distribution

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Pclass

data.boxplot(column = 'Age', by = 'Pclass')
plt.title('Age by Pclass')
plt.suptitle("")
plt.xlabel('Pclass')
plt.ylabel('Age')
plt.show()
```


Output:

CODETANTRA

Home Learn Anywhere

202501040059@mbitce.ac.in Support Logout

5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Age by Pclass"**.
3. Remove the default subtitle with **plt.suptitle('')**.
4. Label the x-axis as **"Pclass"** and the y-axis as **"Age"**.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brandy, Mr. Owen Harris	male	22	1	0	A/5 21171,72,5	5		
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.3833,C85,C			
3	1	3	Helsinki, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.026,5			
4	1	1	Hervella, Mrs. Jacques Heath (Lily May Peel)	female	36	1	0	113883,53.1,C123,5			
5	0	3	Allen, Mr. William Henry	male	35	0	0	37460,8.05,5			
6	0	3	Horn, Mr. James	male	0	0	0	330877,8.4581,0			
7	0	1	McCarthy, Mr. Timothy J.	male	54	0	0	17463,53.8625,646,5			
8	0	3	Paisson, Master. Gosta Leonard	male	2	3	1	349909,21.079,5			
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	1	347142,11.1393,5			
10	1	2	Messer, Mrs. Nicholas (Adeline Acton)	female	18	1	0	237736,30.4708,C			

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlot...

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7
8 # Data Cleaning
9 data['Age'].fillna(data['Age'].median(), inplace=True)
10 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
11 data.drop('Cabin', axis=1, inplace=True)
12
13 # Convert categorical features to numeric
14 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
15 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
16
17 # Write your code here for Box Plot for Age by Pclass
18 data.boxplot(column='Age', by='Pclass')
19 plt.title('Age by Pclass')
20 plt.suptitle('')
21 plt.xlabel('Pclass')
22 plt.ylabel('Age')
23 plt.show()

Average time: 0.227 s
Maximum time: 0.227 s
1 out of 1 shown test case(s) passed

Expected output:

Actual output:

Terminal Test cases

5.2.8 Box Plot for Age by Survived

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).
2. Set the title of the plot to **"Age by Survival"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Age by Survived

plt.figure(figsize = (8,6))
data.boxplot(column = 'Age', by = 'Survived')
plt.title('Age by Survival')
plt.suptitle("")
plt.xlabel('Survived')
plt.ylabel('Age')
plt.show()
```

Output:

CODETANTRA

Home Learn Anywhere

20250104055@mitase.ac.in Support Logout

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the Survived column to group the data for the boxplot (0 = Did not survive, 1 = Survived).

2. Set the title of the plot to "Age by Survival".

3. Remove the default subtitle with plt.suptitle("").

4. Label the x-axis as "Survived" and the y-axis as "Age".

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Mr. Owen Harris	male	22	0	0	15170	7.25	0	0
2	1	1	Mr. John Bradley (Florence Briggs Thayer)	female	38	1	0	17399	53.1	0	0
3	1	3	Mr. Lachlan	male	20	0	0	17402	7.00	0	0
4	1	3	Mr. Joseph Smith (Ellis May Smith)	female	21	0	0	11380	53.1	0	0
5	0	3	Mr. William Henry	male	30	0	0	17418	8.00	0	0
6	0	3	Mr. James	male	26	0	0	15057	8.00	0	0
7	0	3	Mr. James	male	34	0	0	17461	8.00	0	0
8	0	3	Mr. James	male	22	0	0	17469	22.00	0	0
9	1	1	Mr. Oscar W (Elizabeth Villhelmsen Berg)	female	27	0	0	167102	33.00	0	0
10	1	2	Mr. Nicholas	male	34	1	0	237731	38.00	0	0

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1, 0, 3, Mr. Owen Harris, male, 22, 0, 0, 15170, 7.25, 0, 0

2, 1, 1, Mr. John Bradley (Florence Briggs Thayer), female, 38, 1, 0, 17399, 53.1, 0, 0

3, 1, 3, Mr. Lachlan, male, 20, 0, 0, 17402, 7.00, 0, 0

4, 1, 3, Mr. Joseph Smith (Ellis May Smith), female, 21, 0, 0, 11380, 53.1, 0, 0

5, 0, 3, Mr. William Henry, male, 30, 0, 0, 17418, 8.00, 0, 0

6, 0, 3, Mr. James, male, 26, 0, 0, 15057, 8.00, 0, 0

7, 0, 3, Mr. James, male, 34, 0, 0, 17461, 8.00, 0, 0

8, 0, 3, Mr. James, male, 22, 0, 0, 17469, 22.00, 0, 0

9, 1, 1, Mr. Oscar W (Elizabeth Villhelmsen Berg), female, 27, 0, 0, 167102, 33.00, 0, 0

10, 1, 2, Mr. Nicholas, male, 34, 1, 0, 237731, 38.00, 0, 0

Note: Refer to the visible test case for better reference.

Sample Test Cases

BoxPlotForAgeBySurvived

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

23

24

25

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Age by Survived
17 plt.figure(figsize=(8,6))
18 data.boxplot(column='Age', by='Survived')
19
20 plt.title('Age by Survival')
21 plt.suptitle('')
22 plt.xlabel('Survived')
23 plt.ylabel('Age')
24 plt.show()
25
```

0.347 s

247.00 ms

0.347 s

247.00 ms

1 out of 1 shown test case(s) passed

Age by Survival

Age by Survival

Terminal

Test cases

1 Prev

2 Next

5.2.9 Box Plot for Fare by Pclass

Problem Statement:

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass

plt.figure(figsize=(8,6))
data.boxplot(column='Fare', by = 'Pclass')
plt.title('Fare by Pclass')
plt.suptitle("")
plt.xlabel('Pclass')
plt.ylabel('Fare')
plt.show()
```

Output:

CODETANTRAHomeLearn Anywhere202501040059@mitaoe.ac.inSupportLogout

5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **"Pclass"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,7.25,,S			
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.2833,C85,C			
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.925,,S			
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53.1,C123,S			
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8.05,,S			
6	0	3	Moran, Mr. James	male	0	0	0	330877,8.4583,,Q			
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463,51.8625,E46,S			
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	340000,51.070,,C			

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,340000,51.070,,C

BoxPlotF...

1import pandas as pd
2import matplotlib.pyplot as plt
3
4# Load the Titanic dataset
5data = pd.read_csv('Titanic-Dataset.csv')
6
7# Data Cleaning
8data['Age'].fillna(data['Age'].median(), inplace=True)
9data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10data.drop('Cabin', axis=1, inplace=True)
11
12# Convert categorical features to numeric
13data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16# Write your code here for Box Plot for Fare by Pclass
17plt.figure(figsize=(8,6))
18data.boxplot(column='Fare',by='Pclass')
19plt.title("Fare by Pclass")
20plt.suptitle("")
21plt.xlabel('Pclass')
22plt.ylabel("Fare")
23plt.show()
24
25

TerminalTest cases

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5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:
1. Use the **Pclass** column to group the data for the boxplot.
2. Set the title of the plot to **"Fare by Pclass"**.
3. Remove the default subtitle with **plt.suptitle("")**.
4. Label the x-axis as **"Pclass"** and the y-axis as **"Fare"**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171,7.25,,S			
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71.2833,C85,C			
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7.925,,S			
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53.1,C123,S			
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8.05,,S			
6	0	3	Moran, Mr. James	male	0	0	0	330877,8.4583,,Q			
7	0	1	McCarthy, Mr. Timothy J	male	54	0	0	17463,51.8625,E46,S			
8	0	3	Palsson, Master. Gosta Leonard	male	2	3	1	340000,51.070,,C			

Sample Data:
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,340000,51.070,,C

BoxPlotF...

Average time: 0.551 s
Maximum time: 0.551 s
661.00 ms
1 out of 1 shown test case(s) passed

Test case 1 651 ms
Expected output: Fare by Pclass boxplot
Actual output: Fare by Pclass boxplot

TerminalTest cases

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5.2.10 Scatter Plot for Age vs. Fare

Problem Statement:

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Set the title of the plot to **"Age vs. Fare"**.
3. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Box Plot for Fare by Pclass

plt.figure()
plt.scatter(data['Age'], data['Fare'])
plt.title('Age vs. Fare')
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```

Output:

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Logout

6.2.10. Scatter Plot for Age vs. Fare

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

1. Use the Age column for the x-axis and the Fare column for the y-axis.

2. Set the title of the plot to "Age vs. Fare".

3. Label the x-axis as 'Age' and the y-axis as 'Fare'.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,1,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,S

2,1,1,"Cunha, Mrs. John Bradley (Florence Briggs Tayer)",female,36,1,0,PC 17599,71.0033,C&S,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O 3101282,53.1,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,51.1,C323,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,17469,8.05,S

6,0,3,"Moran, Mr. James",male,0,0,330877,0.0001,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,517001,53.0001,S&O,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,104909,21.075,S

9,1,1,"Johnson, Mrs. Oscar W (Ellenath Wilhelmina Hargt)",female,27,0,2,367702,33.3001,S

10,1,2,"Nasser, Mrs. Nicholas (Nahide Houssein)",female,46,1,1,0,210770,30.4700,C

Note: Refer to the visible test case for better reference.

Sample Test Cases

AgeFare...

1 import pandas as pd

2 import matplotlib.pyplot as plt

3

4 # Load the Titanic dataset

5 data = pd.read_csv('Titanic-Dataset.csv')

6

7 # Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Box Plot for Fare by Pclass

17 plt.figure()

18 plt.scatter(data['Age'], data['Fare'])

19 plt.title('Age vs. Fare')

20 plt.xlabel('Age')

21 plt.ylabel('Fare')

22 plt.show()

Average time

0.182 s

182.00 ms

0.182 s

182.00 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Age vs Fare

Age vs Fare

Terminal

Test Cases

1 Prev

2 Next

5.2.11 Scatter Plot for Age vs. Fare by Survived

Problem Statement:

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status.

The scatter plot should display the following specifications:

1. Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
2. Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
3. Set the title of the plot to **"Age vs. Fare by Survival"**.
4. Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')

# Data Cleaning
data['Age'].fillna(data['Age'].median(), inplace=True)
data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
data.drop('Cabin', axis=1, inplace=True)

# Convert categorical features to numeric
data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

# Write your code here for Scatter Plot for Age vs. Fare by Survived

colors = data['Survived'].map({0: 'red', 1: 'blue'})
plt.scatter(data['Age'], data['Fare'], c=colors)

plt.title("Age vs. Fare by Survival")
plt.xlabel('Age')
plt.ylabel('Fare')
plt.show()
```


Output:

5.2.11. Scatter Plot for Age vs. Fare by Survived

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

1. Use the Age column for the x-axis and the Fare column for the y-axis.
2. Color the points based on the Survived column: Red for passengers who did not survive (Survived = 0). Blue for passengers who survived (Survived = 1).
3. Set the title of the plot to "Age vs. Fare by Survival".
4. Label the x-axis as 'Age' and the y-axis as 'Fare'.

The Titanic dataset contains columns as shown below:

Passenger id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,67,2157,7.25,S
2,1,1,"Guthrie, Mrs. James Bradley (Lucy Deane) (Maestri)",female,38,1,0,PC 17599,51.0,0,S
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,31012,53.1,0,S
4,1,1,"Nesbitt, Mrs. Josephine",female,34,1,0,31500,51.0,0,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,37463,8.05,S
6,0,3,"Morris, Mr. James",male,29,0,0,51687,0,0,S
7,0,3,"McCarthy, Mr. Timothy J",male,34,0,0,17461,51.0,0,S
8,0,1,"Palsson, Master. Gosta Leonard",male,2,1,1,34090,21.0,S
9,1,3,"Thomson, Mrs. Oscar W (Elizabeth (Sibbald) Kerr)",female,27,0,2,140792,31.0,S
10,1,1,"Nasser, Mrs. Nicholas (Nadine Akhew)",female,34,1,0,237739,30.0,S
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

Age vs. Fare

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Scatter Plot for Age vs. Fare by Survived
17 colors = data['Survived'].map({0: 'red', 1: 'blue'})
18 plt.scatter(data['Age'], data['Fare'], c=colors)
19 plt.title('Age vs. Fare by Survival')
20 plt.xlabel('Age')
21 plt.ylabel('Fare')
22 plt.show()
```

Average time: 0.216 s, 293.09 ms

Maximum time: 0.216 s, 293.09 ms

1 out of 1 shown test case(s) passed

Test case 1

Expected output

Actual output

Terminal